derive a strategy to adopt emerging technologies to provide more business value than before. Interestingly, security concerns regarding adoption of public cloud providers are falling while those regarding vendor lock-in are increasing. Enterprises do not want to put all their eggs in one cloud provider's basket to minimise their dependency on it.

Important factors that CIOs must consider before promoting a hybrid cloud model across the enterprise are:

- Does the organisation have the right business culture to embrace rapid change?
- What value will the hybrid cloud bring to the business?
- Will it help to lower costs, improve processes and manage security better?
- Is the business ready for the new service model?
- Is it ready to address change management when moving to a hybrid cloud while maintaining business continuity?
- Do enterprises have the right use cases for hybrid cloud adoption? For example, are they using packaged applications, have a disaster recovery setup across multiple geographies, etc.
- What is the volume of requirement for new application development on existing architectures, and for the development of next generation applications?

The hybrid cloud model addresses all these concerns as it offers flexibility, scalability, security, data sovereignty, compliance, and cost-effectiveness based on individual business objectives. Some of the top companies offering hybrid cloud platforms are Amazon Web Services, Google, IBM, Microsoft, Oracle, Salesforce, SAP, Teradata, Alibaba, and Tencent.

The components of a hybrid cloud

The hybrid cloud combines two or more cloud models like on-premises, private cloud, and public cloud, and typically consists of the following (see Figure 1).

Public cloud: Services provided by third-party cloud providers like Amazon Web Services (AWS), Microsoft Azure, or Google Cloud Platform (GCP).

Private cloud or on-premises infrastructure*:* A cloud environment dedicated to a single enterprise, hosted internally or by a third-party provider.

Hybrid cloud management platform: This platform provides tools and services to manage and orchestrate resources across both on-premises and cloud environments. It ensures seamless integration and consistent policies.

Network connectivity: Secure and high-speed connections, such as VPNs or dedicated links like AWS Direct Connect or Azure ExpressRoute, are used to connect on-premises infrastructure with cloud environments.

Security and compliance: Security measures and compliance policies are implemented across all environments to protect data and ensure regulatory adherence.

Key considerations and drivers for moving to a hybrid cloud

A hybrid cloud plays a key role in increasing the speed of delivery of IT resources to end users and improving disaster recovery capabilities while improving resource utilisation.

A few factors that must be considered before moving to a hybrid cloud are listed below.

Current state: The current application landscape and infrastructure across the enterprise must be assessed. A complete application portfolio analysis of the enterprise needs to be carried out.

Portfolio rationalisation: Business functions and the applications that are catering to them must be identified and re-engineered based on industry trends. Redundant functionality across applications should be identified and rationalised before moving on to the hybrid cloud.

Application migration: Applications can be migrated 'to and from' the data centre and cloud using the hybrid cloud model. Identify the applications that need to remain on-premises or move to a private or public cloud. This can be temporary or permanent depending on the strategy of migration.

Nature of applications: Applications that change frequently need to be moved to the cloud to leverage automated deployment through adoption of DevOps. Applications that handle sensitive data are best retained on-premises. Applications with very high scalability requirements because of varied user load should be hosted on the public/private cloud.

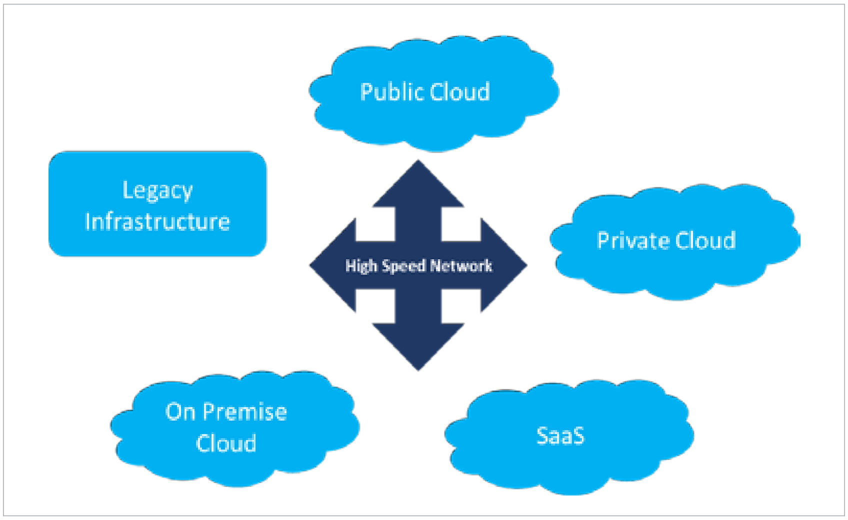

Figure 1: Components of a hybrid cloud